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Bisnis Proses (1 slide , cukup 1 bisnis proses)

Setujui Formulir Rencana Studi (FRS)

What: Dosen wali menyetujui Formulir Rencana Studi (FRS) yang telah diajukan oleh mahasiswa bimbingannya untuk satu semester.

Who:

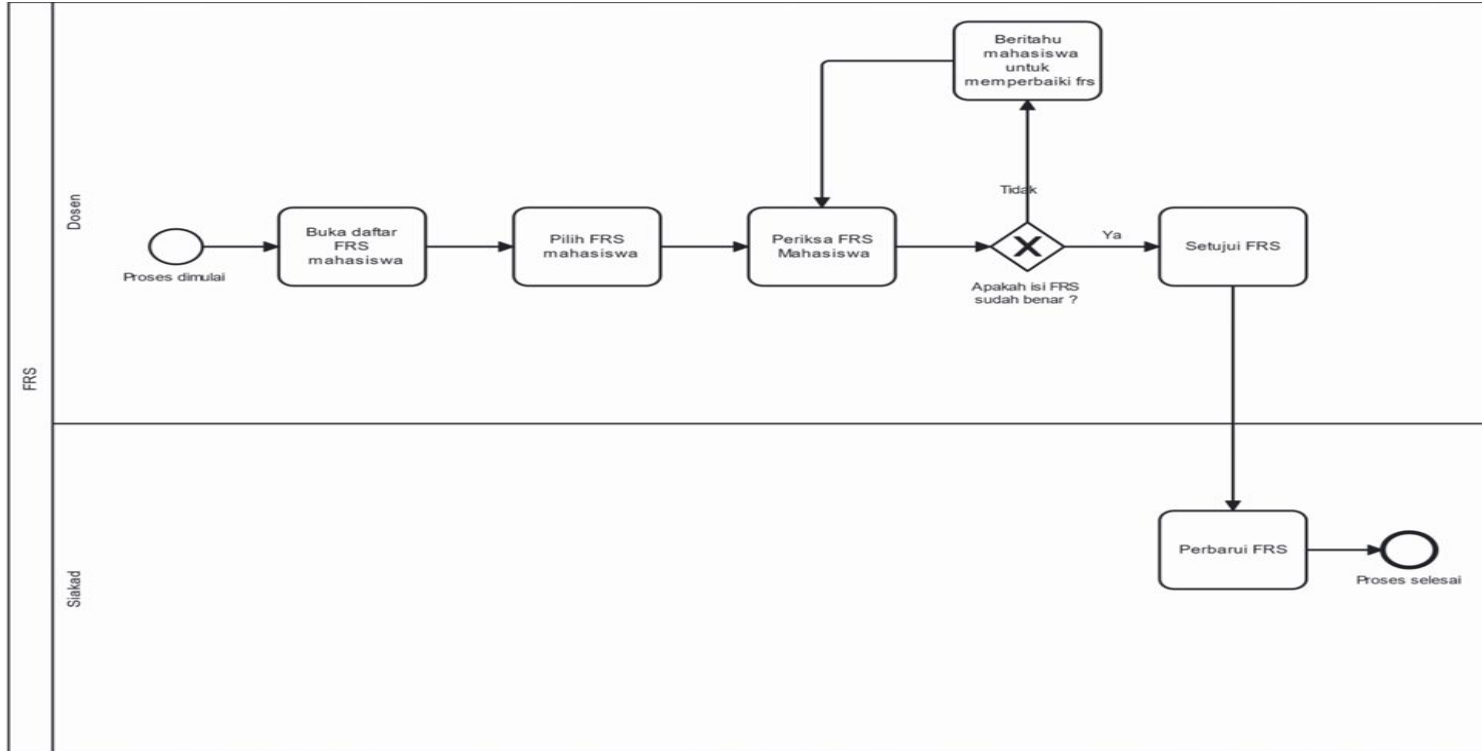
Dosen Wali → Menyetujui atau memberikan catatan pada FRS.

Sistem Akademik → Memvalidasi dan mencatat status persetujuan. When: Pada masa pengisian FRS, menyesuaikan jadwal akademik yang ditetapkan oleh SIAKAD ITS.

Why: Untuk memastikan bahwa mahasiswa memilih mata kuliah sesuai dengan kurikulum, kapasitas akademik, dan kelulusan tepat waktu.

Where: Melalui sistem informasi akademik ITS (SIAKAD), secara online.

How: Mahasiswa mengisi FRS secara online. Setelah itu, dosen wali akan menerima notifikasi, lalu memeriksa dan menyetujui atau memberikan catatan melalui sistem.



Granularitas (1 slide)

Granularitas :

Satu baris dalam tabel fakta = satu persetujuan FRS oleh dosen wali untuk satu mahasiswa pada satu semester.

Dimensi Mahasiswa

Dim_Mahasiswa	
PK	<u>NRP (pk)</u> INT SCD Type 0
	Nama VARCHAR SCD Type 1
	IPK_terakhir Decimal SCD Type 2
	Program_Studi VARCHAR SCD Type 2
	Departemen VARCHAR SCD Type 2
	Jenis kelamin VARCHAR SCD Type 0
	Fakultas VARCHAR SCD Type 1
	Angkatan INT SCD Type 1
	Semester_saat_ini INT SCD Type 1
	Ips_terakhir Decimal SCD Type 2
	Jenjang VARCHAR SCD Type 1
	Tanggal masuk DATE SCD Type 1
	NIP_Dosen VARCHAR SCD Type 2

Dimensi Waktu

Dim_Waktu			
PK	<u>ID_Waktu</u>	INT	SCD Type 0
	Tanggal	DATE	SCD Type 0
	Hari	VARCHAR	SCD Type 0
	Bulan	VARCHAR	SCD Type 0
	Periode	VARCHAR	SCD Type 0

Dimensi Status_permohonan

Dim_Status_Permohonan	
PK	ID_Status (pk) INT SCD Type 0
	Status VARCHAR SCD Type 2

Dimensi Dosen_wali

Dim_Dosen_Wali	
PK	<u>NIP_Dosen(pk)</u> VARCHAR SCD Type 0
	Nama_Dosen VARCHAR SCD Type 1
	Fakultas VARCHAR SCD Type 2
	Departemen VARCHAR SCD Type 2
	Program_Studi VARCHAR SCD Type 2

Fakta (1 slide saja, cukup 1 fakta saja)

- Fakta Jumlah_SKS_FRS | INT | Additive
- Jumlah_MK_FRS | INT | Additive
- Sks tersisa saat ini | INT | Non-additive
- Sks maksimum semester ini | INT | Non-additive

Fact_Setujui	
PK	<u>ID Permohonan</u> INT
FK	<u>NRP</u> INT <u>SCD Type 0</u>
	<u>ID Waktu</u> INT <u>SCD Type 0</u>
	<u>NIP Dosen</u> VARCHAR <u>SCD Type 0</u>
	<u>ID Status</u> INT <u>SCD Type 0</u>
	Jumlah_SKS_FRS INT Additive
	Sks tersisa saat ini INT Non-additive
	Jumlah_MK_FRS INT Additive
	Sks maksimum semester ini INT Non-additive

Desain star schema (1 slide)

Dim_Mahasiswa	
PK	<u>NRP (pk)</u> INT SCD Type 0
	Nama VARCHAR SCD Type 1
	IPK_terakhir Decimal SCD Type 2
	Program_Studi VARCHAR SCD Type 2
	Departemen VARCHAR SCD Type 2
	Jenis kelamin VARCHAR SCD Type 0
	Fakultas VARCHAR SCD Type 1
	Angkatan INT SCD Type 1
	Semester_saat_ini INT SCD Type 1
	Ips_terakhir Decimal SCD Type 2
	Jenjang VARCHAR SCD Type 1
	Tanggal masuk DATE SCD Type 1
	NIP_Dosen VARCHAR SCD Type 2

Fact_Setujui	
PK	<u>ID Permohonan</u> INT
FK	<u>NRP</u> INT SCD Type 0
	<u>ID Waktu</u> INT SCD Type 0
	<u>NIP_Dosen</u> VARCHAR SCD Type 0
	<u>ID Status</u> INT SCD Type 0
	Jumlah_SKS_FRS INT Additive
	Sks tersisa saat ini INT Non-additive
	Jumlah_MK_FRS INT Additive
	Sks maksimum semester ini INT Non-additive

Dim_Status_Permohonan	
PK	<u>ID_Status (pk)</u> INT SCD Type 0
	Status VARCHAR SCD Type 2

Dim_Dosen_Wali	
PK	<u>NIP_Dosen (pk)</u> VARCHAR SCD Type 0
	Nama_Dosen VARCHAR SCD Type 1
	Fakultas VARCHAR SCD Type 2
	Departemen VARCHAR SCD Type 2
	Program_Studi VARCHAR SCD Type 2

Dim_Waktu	
PK	<u>ID Waktu</u> INT SCD Type 0
	Tanggal DATE SCD Type 0
	Hari VARCHAR SCD Type 0
	Bulan VARCHAR SCD Type 0
	Periode VARCHAR SCD Type 0

ETL (1 slide)

<https://youtu.be/NYRQ80cViPI>

eti_db dim_waktu	
❏ waktu_sk : int(11)	
❏ id_waktu : int(11)	
❏ tanggal : date	
❏ tahun : int(11)	
❏ bulan_angka : int(11)	
❏ hari_angka : int(11)	
❏ kuartal : int(11)	
❏ minggu_tahun : int(11)	
❏ hari_dalam_tahun : int(11)	
❏ hari : varchar(20)	
❏ bulan : varchar(20)	
❏ periode : varchar(20)	
❏ tahun_akademik : varchar(9)	
❏ semester : enum('Ganjil','Genap')	
❏ created_date : date	
❏ eti_batch_id : varchar(50)	

eti_db dim_dosen_wali	
❏ dosen_sk : int(11)	
❏ nip_dosen : bigint(20)	
❏ nama_dosen : varchar(100)	
❏ fakultas : varchar(50)	
❏ departemen : varchar(100)	
❏ program_studi : varchar(100)	
❏ has_academic_title : enum('Y','N')	
❏ display_name : varchar(100)	
❏ faculty_department : varchar(150)	
❏ effective_date : date	
❏ end_date : date	
❏ current_flag : enum('Y','N')	
❏ version_number : int(11)	
❏ created_date : date	
❏ updated_date : date	
❏ eti_batch_id : varchar(50)	

eti_db fakta_setujui_frs	
❏ setujui_frs_sk : int(11)	
❏ mahasiswa_sk : int(11)	
❏ waktu_sk : int(11)	
❏ dosen_sk : int(11)	
❏ status_sk : int(11)	
❏ mahasiswa_nk : int(11)	
❏ waktu_nk : int(11)	
❏ dosen_wali_nk : int(11)	
❏ status_permohonan_nk : int(11)	
❏ jumlah_sks_frs : int(11)	
❏ jumlah_mk_frs : int(11)	
❏ sks_tersisa_saas_inis : int(11)	
❏ sks_maksimum_semester_inis : int(11)	
❏ avg_sks_per_mk : decimal(4,2)	
❏ sks_utilization_percentage : decimal(5,2)	
❏ workload_category : enum('Light','Normal','Heavy','Overload')	
❏ data_quality_flag : enum('VALID','INVALID')	
❏ created_date : date	
❏ eti_batch_id : varchar(50)	
❏ record_source : varchar(100)	

eti_db dim_mahasiswa	
❏ mahasiswa_sk : int(11)	
❏ nrp : bigint(20)	
❏ nama : varchar(100)	
❏ ipk_terakhir : decimal(3,2)	
❏ program_studi : varchar(100)	
❏ departemen : varchar(100)	
❏ jenis_kelamin : varchar(20)	
❏ fakultas : varchar(100)	
❏ angkatan : int(11)	
❏ semester_saas_inis : int(11)	
❏ ips_terakhir : decimal(3,2)	
❏ jenjang : varchar(10)	
❏ tanggal_masuk : date	
❏ effective_date : date	
❏ end_date : date	
❏ current_flag : enum('Y','N')	
❏ version_number : int(11)	
❏ created_date : datetime	
❏ updated_date : datetime	
❏ eti_batch_id : varchar(50)	

eti_db dim_status_permohonan	
❏ status_sk : int(11)	
❏ id_status : int(11)	
❏ status : varchar(50)	
❏ status_category : varchar(50)	
❏ status_priority : int(11)	
❏ effective_date : date	
❏ end_date : date	
❏ current_flag : enum('Y','N')	
❏ version_number : int(11)	
❏ created_date : date	
❏ updated_date : date	
❏ eti_batch_id : varchar(50)	

ETL dengan data lake : data bronze (1 slide)

```
C:\Users\micle\Documents\data  buat dlh\duckdb>duckdb frs_lakehouse.duckdb < etl_pipeline_duckdb.sql

C:\Users\micle\Documents\data  buat dlh\duckdb>duckdb frs_lakehouse.duckdb
DuckDB v1.3.1 (Ossivalis) 2063dda3e6
Enter ".help" for usage hints.
```

```
CREATE OR REPLACE TABLE bronze_mahasiswa AS
SELECT * FROM read_csv_auto('C:/Users/micel/Documents/data  buat dlh/duckdb/Dim_Mahasiswa.csv', HEADER=TRUE, AUTO_DETECT=TRUE);
CREATE OR REPLACE TABLE bronze_dosen_wali AS
SELECT * FROM read_csv_auto('C:/Users/micel/Documents/data  buat dlh/duckdb/Dim_Dosen_Wali (3).csv', HEADER=TRUE, AUTO_DETECT=TRUE);
CREATE OR REPLACE TABLE bronze_status_permohonan AS
SELECT * FROM read_csv_auto('C:/Users/micel/Documents/data  buat dlh/duckdb/Dim_Status_Permohonan (2).csv', HEADER=TRUE, AUTO_DETECT=TRUE);
CREATE OR REPLACE TABLE bronze_waktu AS
SELECT * FROM read_csv_auto('C:/Users/micel/Documents/data  buat dlh/duckdb/Dim_Waktu (2).csv', HEADER=TRUE, AUTO_DETECT=TRUE);
CREATE OR REPLACE TABLE bronze_setujui_frs AS
SELECT * FROM read_csv_auto('C:/Users/micel/Documents/data  buat dlh/duckdb/Setujui_FRS.csv', HEADER=TRUE, AUTO_DETECT=TRUE);
```

ETL dengan data lake : data silver (1 slide)

```
CREATE OR REPLACE TABLE silver_dim_waktu AS
SELECT
  row_number() OVER () AS waktu_sk,
  ID_Waktu AS id_waktu,
  TRY_CAST(Tanggal AS DATE) AS tanggal,
  EXTRACT(year FROM TRY_CAST(Tanggal AS DATE)) AS tahun,
  EXTRACT(month FROM TRY_CAST(Tanggal AS DATE)) AS bulan_angka,
  EXTRACT(day FROM TRY_CAST(Tanggal AS DATE)) AS hari_angka,
  CEIL(EXTRACT(month FROM TRY_CAST(Tanggal AS DATE))/3) AS kuartal,
  strftime(TRY_CAST(Tanggal AS DATE), '%u')::INTEGER AS minggu_tahun,
  (TRY_CAST(Tanggal AS DATE) - DATE_TRUNC('year', TRY_CAST(Tanggal AS DATE)) + 1)::INTEGER AS hari_dalam_tahun,
  upper(substr(Hari, 1, 1)) || lower(substr(Hari, 2)) AS hari,
  upper(substr(Bulan, 1, 1)) || lower(substr(Bulan, 2)) AS bulan,
  Periode,
  TRY_CAST(SPLIT_PART(Periode, '-', 1) AS INTEGER) AS tahun_akademik,
  CASE WHEN SPLIT_PART(Periode, '-', 2) = '1' THEN 'Ganjil' WHEN SPLIT_PART(Periode, '-', 2) = '2' THEN 'Genap' ELSE '' END AS semester,
  CURRENT_DATE AS created_date,
  'BATCH_' || strftime(CURRENT_DATE, '%Y%m%d') || '_001' AS etl_batch_id
FROM bronze_waktu;
```

```
CREATE OR REPLACE TABLE silver_dim_dosen_wali AS
SELECT
  row_number() OVER () AS dosen_sk,
  TRY_CAST(NIP_Dosen AS BIGINT) AS nip_dosen,
  TRIM>Nama_Dosen AS nama_dosen,
  TRIM(Fakultas) AS fakultas,
  TRIM(Departemen) AS departemen,
  TRIM(Program_Studi) AS program_studi,
  CASE WHEN Nama_Dosen ~ 'Dr\\.|Prof\\.|Ir\\.|S.T\\.\\.|M.T\\.\\.|M.Sc\\.\\.|Ph.D.' THEN 'Y' ELSE 'N' END AS has_academic_title,
  regexp_replace>Nama_Dosen, 'Dr\\.|Prof\\.|Ir\\.|S.T\\.\\.|M.T\\.\\.|M.Sc\\.\\.|Ph.D.', '', 'g') AS display_name,
  CASE WHEN Fakultas IS NOT NULL AND Departemen IS NOT NULL THEN Fakultas || '-' || Departemen
  WHEN Fakultas IS NOT NULL THEN Fakultas
  ELSE Departemen END AS faculty_department,
  CURRENT_DATE AS effective_date,
  DATE '9999-12-31' AS end_date,
  'Y' AS current_flag,
  1 AS version_number,
  CURRENT_DATE AS created_date,
  CURRENT_DATE AS updated_date,
  'BATCH_' || strftime(CURRENT_DATE, '%Y%m%d') || '_001' AS etl_batch_id
FROM bronze_dosen_wali;
```

```
CREATE OR REPLACE TABLE silver_dim_mahasiswa AS
SELECT
  row_number() OVER () AS mahasiswa_sk,
  CAST(MRP AS BIGINT) AS mnp,
  TRIM>Nama AS nama,
  TRY_CAST(ROUND(CAST(IPK_terakhir AS DOUBLE),2) AS DOUBLE) AS ipk_terakhir,
  TRIM(Program_Studi) AS program_studi,
  TRIM(Departemen) AS departemen,
  CASE
    WHEN lower("jenis_kelamin") LIKE 'laki-laki' THEN 'Laki-laki'
    WHEN lower("jenis_kelamin") LIKE 'perempuan' OR lower("jenis_kelamin") LIKE 'wanita' THEN 'Perempuan'
    ELSE TRIM("jenis_kelamin")
  END AS jenis_kelamin,
  TRIM(Fakultas) AS fakultas,
  TRY_CAST(Angkatan AS INTEGER) AS angkatan,
  CASE WHEN TRY_CAST(Semester_saai_in AS INTEGER) BETWEEN 1 AND 14 THEN TRY_CAST(Semester_saai_in AS INTEGER) ELSE NULL END AS semester_saai_in,
  TRY_CAST(ROUND(CAST(IPK_terakhir AS DOUBLE),2) AS DOUBLE) AS ipk_terakhir,
  CASE WHEN upper(Cepiting) IN ('S1','S2','S3') THEN upper(Jenjang) ELSE Jenjang END AS jenjang,
  TRY_CAST(Tanggal_masuk AS DATE) AS tanggal_masuk,
  CURRENT_DATE AS effective_date,
  DATE '9999-12-31' AS end_date,
  'Y' AS current_flag,
  1 AS version_number,
  CURRENT_DATE AS created_date,
  CURRENT_DATE AS updated_date,
  'BATCH_' || strftime(CURRENT_DATE, '%Y%m%d') || '_001' AS etl_batch_id
FROM bronze_mahasiswa;
```

```
CREATE OR REPLACE TABLE silver_fakta_setujui_frs AS
SELECT
  row_number() OVER () AS setujuifrs_sk,
  m.mahasiswa_sk,
  w.waktu_sk,
  d.dosen_sk,
  s.status_sk,
  TRY_CAST(b.Mahasiswa_SK AS BIGINT) AS mahasiswa_nk,
  TRY_CAST(b.Waktu_SK AS BIGINT) AS waktu_nk,
  TRY_CAST(b.Dosen_Wali_SK AS BIGINT) AS dosen_wali_nk,
  TRY_CAST(b.Status_Permohonan_SK AS BIGINT) AS status_permohonan_nk,
  TRY_CAST(b.Jumlah_SKS_FRS AS INTEGER) AS jumlah_sks_frs,
  TRY_CAST(b.Jumlah_MK_FRS AS INTEGER) AS jumlah_mk_frs,
  TRY_CAST(b.Sks_tersisa_saat_in AS INTEGER) AS sks_tersisa_saat_in,
  TRY_CAST(b.Sks_maksimum_semester_in AS INTEGER) AS sks_maksimum_semester_in,
  CASE WHEN TRY_CAST(b.Jumlah_MK_FRS AS INTEGER) > 0 THEN ROUND(CAST(b.Jumlah_SKS_FRS AS DOUBLE)/CAST(b.Jumlah_MK_FRS AS DOUBLE),2) ELSE NULL END AS avg_sks_per_mk,
  CASE WHEN TRY_CAST(b.Sks_maksimum_semester_in AS INTEGER) > 0 THEN ROUND(CAST(b.Jumlah_SKS_FRS AS DOUBLE)/CAST(b.Sks_maksimum_semester_in AS DOUBLE)*100,2) ELSE NULL END AS avg_sks_per_mk,
  CASE WHEN TRY_CAST(b.Jumlah_SKS_FRS AS INTEGER) < 12 THEN 'Light'
  WHEN TRY_CAST(b.Jumlah_SKS_FRS AS INTEGER) <= 18 THEN 'Normal'
  WHEN TRY_CAST(b.Jumlah_SKS_FRS AS INTEGER) > 18 THEN 'Heavy'
  ELSE 'Unknown' END AS workload_category,
  'VALID' AS data_quality_flag,
  CURRENT_DATE AS created_date,
  'BATCH_' || strftime(CURRENT_DATE, '%Y%m%d') || '_001' AS etl_batch_id,
  'setujuiFRS.csv' AS record_source
FROM bronze_setujui_frs b
LEFT JOIN silver_dim_mahasiswa m ON m.mnp = TRY_CAST(b.Mahasiswa_SK AS BIGINT)
LEFT JOIN silver_dim_waktu w ON w.id_waktu = TRY_CAST(b.Waktu_SK AS BIGINT)
LEFT JOIN silver_dim_dosen_wali d ON d.nip_dosen = TRY_CAST(b.Dosen_Wali_SK AS BIGINT)
LEFT JOIN silver_dim_status_permohonan s ON s.id_status = TRY_CAST(b.Status_Permohonan_SK AS BIGINT);
```

```
CREATE OR REPLACE TABLE silver_dim_status_permohonan AS
SELECT
  row_number() OVER () AS status_sk,
  ID_Status AS id_status,
  TRIM>Status AS status,
  CASE
    WHEN lower(Status) IN ('disetujui','selesai') THEN 'Approved'
    WHEN lower(Status) IN ('ditolak','dibatalkan') THEN 'Rejected'
    WHEN lower(Status) IN ('diproses','pending') THEN 'In Progress'
    ELSE 'Other' END AS status_category,
  CASE
    WHEN lower(Status) = 'pending' THEN 1
    WHEN lower(Status) = 'diproses' THEN 2
    WHEN lower(Status) = 'disetujui' THEN 3
    WHEN lower(Status) = 'selesai' THEN 4
    WHEN lower(Status) = 'ditolak' THEN 5
    WHEN lower(Status) = 'dibatalkan' THEN 6
    ELSE 99 END AS status_priority,
  CURRENT_DATE AS effective_date,
  DATE '9999-12-31' AS end_date,
  'Y' AS current_flag,
  1 AS version_number,
  CURRENT_DATE AS created_date,
  CURRENT_DATE AS updated_date,
  'BATCH_' || strftime(CURRENT_DATE, '%Y%m%d') || '_001' AS etl_batch_id
FROM bronze_status_permohonan;
```

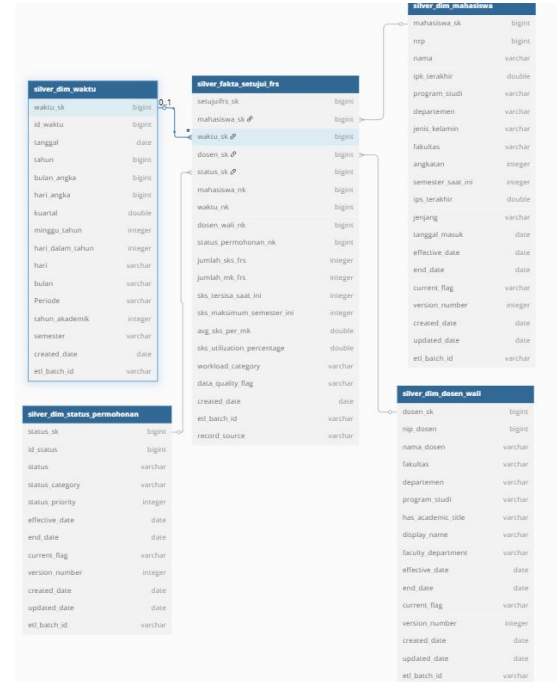

ETL dengan data lake : data gold (1 slide)

```

COPY silver_dim_mahasiswa TO 'C:/Users/micle/Documents/data buat dlh/duckdb/gold/dim_mahasiswa.parquet' (FORMAT PARQUET);
COPY silver_dim_dosen_wali TO 'C:/Users/micle/Documents/data buat dlh/duckdb/gold/dim_dosen_wali.parquet' (FORMAT PARQUET);
COPY silver_dim_status_permohonan TO 'C:/Users/micle/Documents/data buat dlh/duckdb/gold/dim_status_permohonan.parquet' (FORMAT PARQUET);
COPY silver_dim_waktu TO 'C:/Users/micle/Documents/data buat dlh/duckdb/gold/dim_waktu.parquet' (FORMAT PARQUET);
COPY silver_fakta_setujui_frs TO 'C:/Users/micle/Documents/data buat dlh/duckdb/gold/fakta_setujui_frs.parquet' (FORMAT PARQUET);
    
```

```

D. schema
CREATE TABLE bronze_dosen_wali(NIP_Dosen BIGINT, Nama_Dosen VARCHAR, Fakultas VARCHAR, Departemen VARCHAR, Program_Studi VARCHAR);
CREATE TABLE bronze_mahasiswa(NRP BIGINT, Nama VARCHAR, IPK_terakhir DOUBLE, Program_Studi VARCHAR, Departemen VARCHAR, "Jenis_kelamin" VARCHAR, Fakultas VARCHAR, Angkatan BIGINT, Semester_saat_ini BIGINT, Ips_terakhir DOUBLE, Jenjang_V
RCHAR, "Tanggal_masuk" DATE);
CREATE TABLE bronze_setujui_frs(Mahasiswa_SK BIGINT, Waktu_SK BIGINT, Status_Pemohonan_SK BIGINT, Dosen_Wali_SK BIGINT, Jumlah_SKS_FRS BIGINT, Jumlah_MK_FRS BIGINT, Sks_tersisa_saat_ini BIGINT, Sks_maksimum_semester_ini BIGINT);
CREATE TABLE bronze_status_permohonan(ID_Status BIGINT, Status VARCHAR);
CREATE TABLE bronze_waktu(ID_Waktu BIGINT, Tanggal DATE, Hari VARCHAR, Bulan VARCHAR, Periode VARCHAR);
CREATE TABLE silver_dim_dosen_wali(dosen_sk BIGINT, nip_dosen BIGINT, nama_dosen VARCHAR, fakultas VARCHAR, departemen VARCHAR, program_studi VARCHAR, has_academic_title VARCHAR, display_name VARCHAR, faculty_department VARCHAR, effecti
e DATE, end_date DATE, current_flag VARCHAR, version_number INTEGER, created_date DATE, updated_date DATE, etl_batch_id VARCHAR);
CREATE TABLE silver_dim_mahasiswa(mahasiswa_sk BIGINT, nrp BIGINT, nama VARCHAR, ipk_terakhir DOUBLE, program_studi VARCHAR, departemen VARCHAR, jenis_kelamin VARCHAR, fakultas VARCHAR, angkatan INTEGER, semester_saat_ini INTEGER, ips_t
rakhir DOUBLE, jenjang VARCHAR, tanggal_masuk DATE, effective_date DATE, end_date DATE, current_flag VARCHAR, version_number INTEGER, created_date DATE, updated_date DATE, etl_batch_id VARCHAR);
CREATE TABLE silver_dim_status_permohonan(status_sk BIGINT, id_status BIGINT, status VARCHAR, status_category VARCHAR, status_priority INTEGER, effective_date DATE, end_date DATE, current_flag VARCHAR, version_number INTEGER, created_d
e DATE, updated_date DATE, etl_batch_id VARCHAR);
CREATE TABLE silver_dim_waktu(waktu_sk BIGINT, id_waktu BIGINT, tanggal DATE, tahun BIGINT, bulan_angka BIGINT, hari_angka BIGINT, kuartal DOUBLE, minggu_tahun INTEGER, hari_dalam_tahun INTEGER, hari VARCHAR, bulan VARCHAR, Periode V
RCHAR, tahun_akademik INTEGER, semester VARCHAR, created_date DATE, etl_batch_id VARCHAR);
CREATE TABLE silver_fakta_setujui_frs(setujui_frs_sk BIGINT, mahasiswa_sk BIGINT, waktu_sk BIGINT, dosen_sk BIGINT, status_sk BIGINT, mahasiswa_nk BIGINT, waktu_nk BIGINT, dosen_wali_nk BIGINT, status_permohonan_nk BIGINT, jumlah_sks_frs
INTEGER, jumlah_mk_frs INTEGER, sks_tersisa_saat_ini INTEGER, sks_maksimum_semester_ini INTEGER, avg_sks_per_mk DOUBLE, sks_utilization_percentage DOUBLE, workload_category VARCHAR, data_quality_flag VARCHAR, created_date DATE, etl_batc
h_id VARCHAR, record_source VARCHAR);
    
```



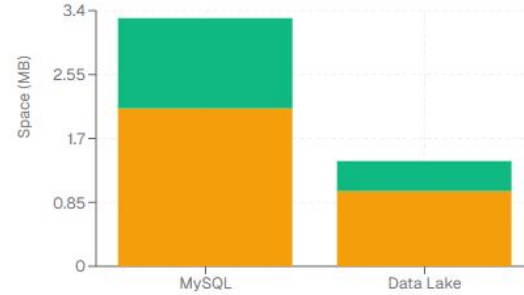
Komparasi hasil antara MySQL dan DataLake (1 Slide)

Waktu etl



Space

Hijau = final | Orange = temp



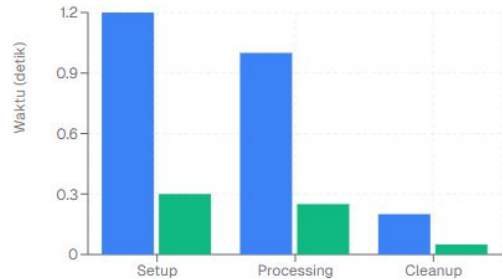
Waktu proses per tabel

Biru = Mysql | hijau = datalake



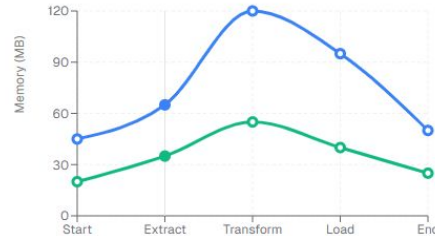
Per phase etl

Biru = Mysql | hijau = datalake



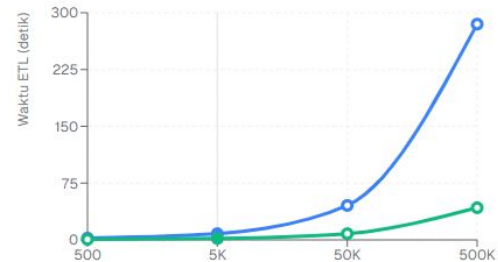
Memory

Biru = Mysql | hijau = datalake



Scalability dataset

Biru = Mysql | hijau = datalake



Ujicoba Starschema

Link Youtube :

<https://youtu.be/XGodHj0i4TQ>

Renungan

Saya memperoleh banyak wawasan dan keterampilan teknis yang sebelumnya belum saya kuasai. Beberapa hal penting yang saya pelajari meliputi:

- Pemahaman konsep Data Warehouse dan Data Lake, termasuk perbedaan antara pendekatan tradisional dan modern dalam manajemen data besar.
- Desain skema bintang (star schema) sebagai pendekatan dimensional modeling yang efektif untuk analisis data, serta pentingnya granularitas dalam perancangan tabel fakta.
- Penerapan proses ETL (Extract, Transform, Load) baik menggunakan pendekatan konvensional dengan MySQL maupun modern dengan Data Lake (bronze, silver, gold layer).
- Penggunaan tools seperti DuckDB, dan Pentaho